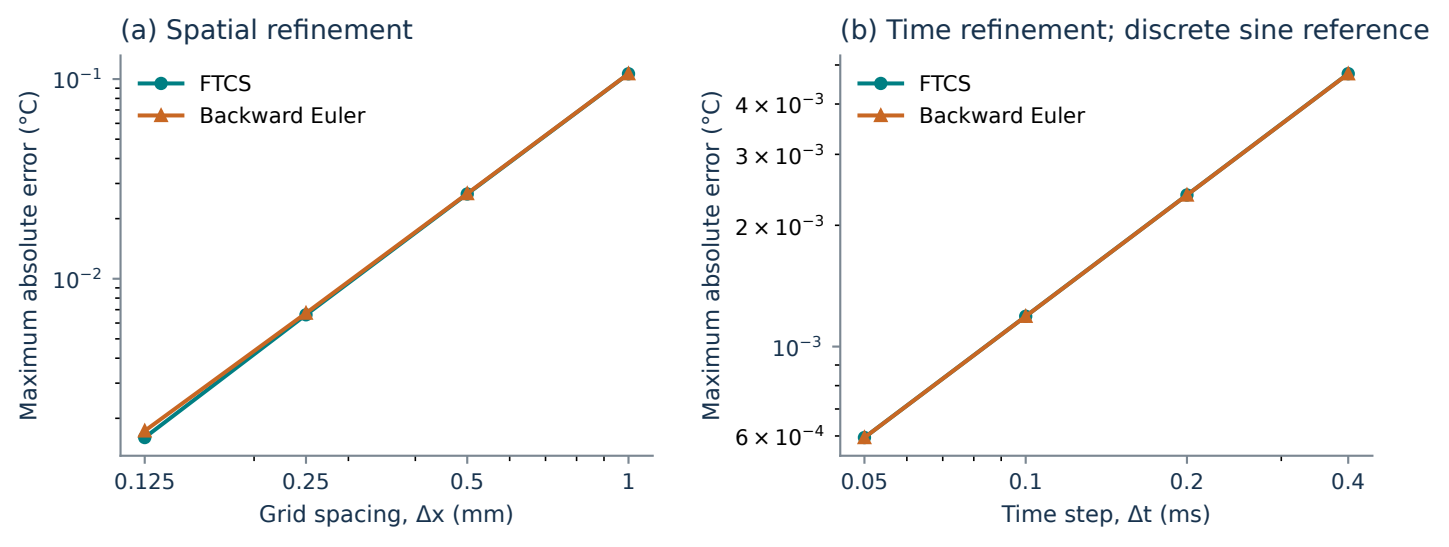
(a) Spatial refinement
FTCS
Backward Euler
Maximum absolute error (°C)
10^{-1}
10^{-2}
0.125
0.25
0.5
1
Grid spacing, Δx (mm)
(b) Time refinement; discrete sine reference
FTCS
Backward Euler
Maximum absolute error (°C)
4×10^{-3}
3×10^{-3}
2×10^{-3}
10^{-3}
6×10^{-4}
0.05
0.1
0.2
0.4
Time step, Δt (ms)